

Precalc Lesson 1.1 Homework

Name _____

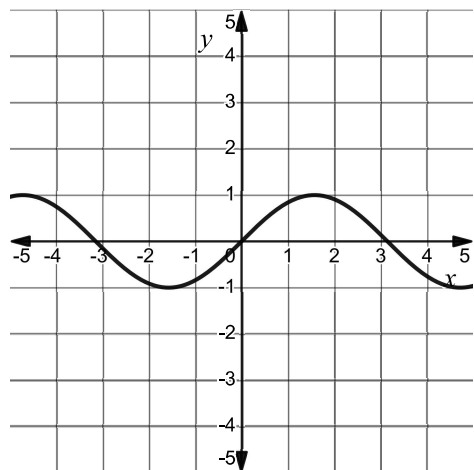
1. Determine if each of the following relationships can be represented with a function f where $f(x) = y$. Give a reason for your answer.

a. A child's height in inches, y , on their birthday at age x

b. x = month of the year, y = average weekly price of gas at a gas station

c. The cost of mailing a small package, y , based on its weight, x

d.



e. $\{ (3, 2), (-1, 8), (5, 8), (-2, -9), (-1, 10) \}$

2. Let $h(x) = x^2 - 3x + 6$. Find the following values:

a. $h(3)$

b. $h(-2)$

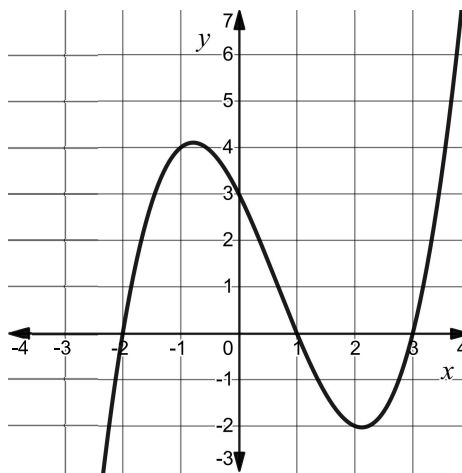
3. A table of selected values is given for a function f .

x	$f(x)$
-5	11
-2	8
0	-3
2	4
7	-2

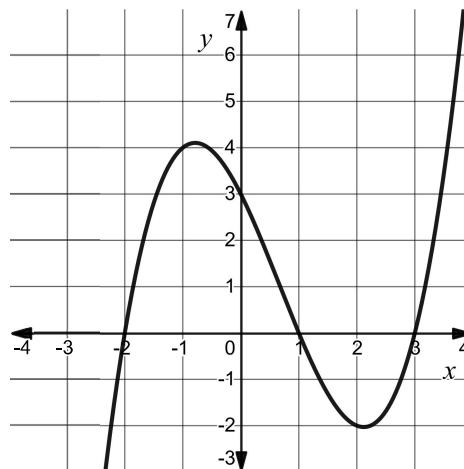
a. Find $f(2)$.

b. What is the value of x when $f(x) = -2$?

4. The graph of $y = g(x)$ is shown. Find $g(-1)$.

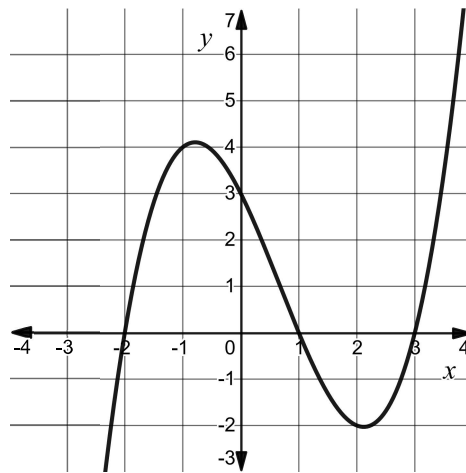


5. The graph of $y = g(x)$ is shown. For which value(s) of x is $g(x) = 0$?



6. The graph of $y = g(x)$ is shown. Order the following from least to greatest:

$$g(-1), g(-2), g(0.5), g(2.5)$$



7. Let $M(t)$ represent the rate at which people enter Macy's department store, in people per hour, t hours into their Black Friday sale. Interpret the statement $M(3.5) = 432$ in the context of this problem.
8. Let $L(t)$ represent the number of likes on a social media post, t hours after it is posted. Interpret the statement $L(4) = 2L(3)$.
9. Describe a relationship between two variables, x and y , that would NOT be considered a function. In your description, be sure to define the two variables and provide any units, and explain why the relationship is not a function.